

Midterm 1 – Solutions

ECON 301
Spring 2021



Question 1

(a) We need first the conditional pdf $f(y|x) = \frac{f(x,y)}{f(x)}$

$$\begin{aligned} f(x) &= \int_{-\infty}^{\infty} \frac{3}{11} (x^2 + y) 1_{[0,2]}(x) \cdot 1_{[0,1]}(y) dy = \int_0^1 \frac{3}{11} (x^2 + y) 1_{[0,2]}(x) dy \\ &= \frac{3}{11} 1_{[0,2]}(x) \int_0^1 (x^2 + y) dy = \frac{3}{11} 1_{[0,2]}(x) \left[x^2 y + \frac{y^2}{2} \right]_0^1 = \frac{3}{11} 1_{[0,2]}(x) \left(x^2 + \frac{1}{2} \right) \\ &= \frac{3}{22} (2x^2 + 1) 1_{[0,2]}(x) \end{aligned}$$

$$\text{Thus, } f(y|x) = \frac{\frac{3}{11} (x^2 + y) 1_{[0,2]}(x) \cdot 1_{[0,1]}(y)}{\frac{3}{22} (2x^2 + 1) 1_{[0,2]}(x)} = 2 \left(\frac{x^2 + y}{2x^2 + 1} \right) 1_{[0,1]}(y) \text{ whenever } 0 \leq x \leq 2$$

Hence $E[Y|X=x]$ will only be defined for $x \in [0,2]$. We have that

$$\begin{aligned} E[Y|X=x] &= \int_{-\infty}^{\infty} y f(y|x) dy = \int_{-\infty}^{\infty} y \left\{ 2 \left(\frac{x^2 + y}{2x^2 + 1} \right) 1_{[0,1]}(y) \right\} dy = \int_0^1 \left(\frac{2}{2x^2 + 1} \right) (x^2 y + y^2) dy \\ &= \frac{2}{2x^2 + 1} \left[x^2 \frac{y^2}{2} + \frac{y^3}{3} \right]_0^1 = \left(\frac{2}{2x^2 + 1} \right) \left(\frac{x^2}{2} + \frac{1}{3} \right) = \left(\frac{2}{2x^2 + 1} \right) \left(\frac{3x^2 + 2}{6} \right) = \frac{1}{3} \left(\frac{3x^2 + 2}{2x^2 + 1} \right) \end{aligned}$$

(b) False since the CEF is not linear

(c) The BLP is given by $E^*(Y|X) = \alpha + \beta X$ where $\alpha = E(Y) - \beta E(X)$, $\beta = \frac{\text{Cov}(X,Y)}{\text{Var}(X)}$

Thus, we need to compute the following moments: $E(X)$, $E(Y)$, $E(X^2)$, $E(XY)$

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x \frac{3}{22} (2x^2+1) 1_{[0,2]}(x) dx = \frac{3}{22} \int_0^2 (2x^3+x) dx = \frac{3}{22} \left(2 \frac{x^4}{4} + \frac{x^2}{2} \right)_0^2 = \frac{3}{22} \left(\frac{1}{2}(16) + \frac{4}{2} \right) \\ &= \frac{3}{22} (8+2) = \frac{30}{22} = \frac{15}{11} \end{aligned}$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 \frac{3}{22} (2x^2+1) 1_{[0,2]}(x) dx = \frac{3}{22} \int_0^2 (2x^4+x^2) dx = \frac{3}{22} \left(2 \frac{x^5}{5} + \frac{x^3}{3} \right)_0^2 = \frac{3}{22} \left(\frac{2}{5}(32) + \frac{8}{3} \right) \\ &= \frac{3}{22} \left(\frac{192+40}{15} \right) = \frac{696}{330} = \frac{116}{55} \end{aligned}$$

$$\therefore \text{Var}(X) = \frac{116}{55} - \left(\frac{15}{11} \right)^2 = \frac{116}{55} - \frac{225}{121} = \frac{151}{605}$$

$$\begin{aligned} E(Y) &= E[E(Y|X)] = \int_{-\infty}^{\infty} \left[\frac{1}{3} \left(\frac{3x^2+2}{2x^2+1} \right) \right] \left[\frac{3}{22} (2x^2+1) 1_{[0,2]}(x) \right] dx = \frac{1}{22} \int_0^2 (3x^2+2) dx = \frac{1}{22} \left(\frac{3x^3}{3} + 2x \right)_0^2 \\ &= \frac{1}{22} (8+4) = \frac{12}{22} = \frac{6}{11} \end{aligned}$$

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \left(\frac{3}{11} (x^2+y) 1_{[0,2]}(x) 1_{[0,1]}(y) \right) dy dx = \frac{3}{11} \int_0^2 \int_0^1 (x^3y + xy^2) dy dx = \frac{3}{11} \int_0^2 \left(x^3 \frac{y^2}{2} + x \frac{y^3}{3} \right)_0^1 dx$$

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$$= \frac{3}{11} \int_0^2 \left(\frac{x^3}{2} + \frac{x}{3} \right) dx = \frac{3}{11} \cdot \frac{1}{6} \int_0^2 (3x^3 + 2x) dx = \frac{1}{22} \left(\frac{3x^4}{4} + \frac{2x^2}{2} \right)_0^2 = \frac{1}{22} \left(\frac{3}{4}(16) + 4 \right) = \frac{8}{11}$$

$$\therefore \text{Cov}(X, Y) = \frac{8}{11} - \frac{15}{11} \cdot \frac{6}{11} = \frac{88 - 90}{121} = -\frac{2}{121}$$

$$\text{Hence, } \beta = \frac{-2/121}{151/605} = -\frac{(2)(605)}{(151)(121)} = -\frac{(2)(5)}{151} = -\frac{10}{151} \approx -0.0662$$

$$\alpha = \frac{6}{11} - \left(-\frac{10}{151} \right) \left(\frac{15}{11} \right) = \frac{1}{11} \left(6 + \frac{150}{151} \right) = \frac{1}{11} \left(\frac{1056}{151} \right) = \frac{96}{151} \approx 0.6358$$

$$\text{Therefore, } E^*(Y|X) = \frac{96}{151} - \frac{10}{151} X \approx 0.6358 - 0.0662X$$

Question 2

$$(a) \hat{\delta} = \underset{d}{\operatorname{argmin}} \sum_{i=1}^n (Y_i - dX_i)^2$$

The first-order condition is: $-2 \sum_{i=1}^n X_i (Y_i - \hat{\delta} X_i) = 0 \Leftrightarrow \sum_{i=1}^n X_i (Y_i - \hat{\delta} X_i) = 0 \Leftrightarrow$

$$\Leftrightarrow \sum_{i=1}^n X_i Y_i = \hat{\delta} \sum_{i=1}^n X_i^2. \quad \text{Thus } \boxed{\hat{\delta} = \frac{\sum X_i Y_i}{\sum X_i^2}}. \quad \text{Notice that A3' implies } \hat{\delta} \text{ exists}$$

and is unique.

$$(b) E(\hat{\delta} | X_s) = E\left(\frac{\sum X_i Y_i}{\sum X_i^2} \mid X_s\right) = \frac{1}{\sum X_i^2} E\left[\sum X_i Y_i \mid X_s\right] = \frac{1}{\sum X_i^2} \sum E(X_i Y_i \mid X_s)$$

$$= \frac{1}{\sum X_i^2} \sum X_i E(Y_i \mid X_s) = \frac{1}{\sum X_i^2} \sum X_i E(Y_i \mid X_i) = \frac{1}{\sum X_i^2} \sum X_i \delta X_i = \delta \frac{\sum X_i^2}{\sum X_i^2} = \delta$$

$\uparrow$ by strong exogeneity $\uparrow$ by A2'

$\therefore \hat{\delta}$ is conditionally unbiased for δ .

$$\operatorname{Var}(\hat{\delta} | X_s) = \operatorname{Var}\left(\frac{\sum X_i Y_i}{\sum X_i^2} \mid X_s\right) = \frac{1}{(\sum X_i^2)^2} \operatorname{Var}\left(\sum X_i Y_i \mid X_s\right) = \frac{1}{(\sum X_i^2)^2} \sum \operatorname{Var}(X_i Y_i \mid X_s)$$

$$\stackrel{\text{by strong exogeneity again}}{=} \frac{1}{(\sum X_i^2)^2} \sum X_i^2 \operatorname{Var}(Y_i \mid X_s) \stackrel{\text{by strong exogeneity}}{=} \frac{1}{(\sum X_i^2)^2} \sum X_i^2 \operatorname{Var}(Y_i \mid X_i) \stackrel{\text{by A4}}{=} \frac{1}{(\sum X_i^2)^2} \sum X_i^2 \sigma^2 = \frac{\sigma^2}{\sum X_i^2}$$

$$(c) E(\hat{\beta}_1 | X's) = E\left(\frac{S_{xy}}{S_{xx}} | X's\right) = \frac{1}{S_{xx}} E\left(\sum_i (x_i - \bar{x}) y_i | X's\right) = \frac{1}{S_{xx}} \sum_i (x_i - \bar{x}) E(y_i | X's)$$

by Lemma 3.1

$$= \frac{1}{S_{xx}} \sum_i (x_i - \bar{x}) E(y_i | x_i) = \frac{1}{S_{xx}} \sum_i (x_i - \bar{x}) \delta x_i = \delta \frac{S_{xx}}{S_{xx}} = \delta$$

by strong exogeneity

by Lemma 3.1

$\therefore \hat{\beta}_1$ is also conditionally unbiased for δ .

$$\text{Var}(\hat{\beta}_1 | X's) = \text{Var}\left(\frac{S_{xy}}{S_{xx}} | X's\right) = \frac{1}{S_{xx}^2} \text{Var}\left(\sum_i (x_i - \bar{x}) y_i | X's\right) = \frac{1}{S_{xx}^2} \sum_i (x_i - \bar{x})^2 \text{Var}(y_i | X's)$$

$$\stackrel{\text{by strong exogeneity}}{=} \frac{1}{S_{xx}^2} \sum_i (x_i - \bar{x})^2 \text{Var}(y_i | x_i) = \frac{1}{S_{xx}^2} \sum_i (x_i - \bar{x})^2 \sigma^2 = \frac{\sigma^2}{S_{xx}^2} S_{xx} = \frac{\sigma^2}{S_{xx}}$$

by strong exogeneity

by A4

(d) True. Both $\hat{\delta}$ and $\hat{\beta}$ are conditionally unbiased for δ . Also note that

$$\text{Var}(\hat{\delta} | X's) \leq \text{Var}(\hat{\beta} | X's) \Leftrightarrow \frac{\sigma^2}{\sum_i x_i^2} \leq \frac{\sigma^2}{S_{xx}} \Leftrightarrow \sum_i x_i^2 \geq S_{xx} \Leftrightarrow \sum_i x_i^2 \geq \sum_i x_i^2 - n\bar{x}^2$$

(This last equivalence comes from Lemma 3.1)

Since by $\sum_i x_i^2 > 0$ and $\bar{x}^2 \geq 0$, the last statement is true.

Question 3

(a) By A2, the model is $E(\text{wage} | IQ) = \beta_0 + \beta_1 IQ$. Thus $\frac{\partial E(\text{wage} | IQ)}{\partial IQ} = \beta_1$.

$\therefore \beta_1$ is the ceteris paribus effect of IQ on the conditional expected wage.

The interpretation would be that one extra IQ point increases wage by β_1 monetary units. This effect is estimated to be $\hat{\beta}_1 = 8.3031$.

(b) The hypotheses to test are in mathematical terms: $H_0: \beta_1 = 10$ vs $H_a: \beta_1 > 10$

The test statistic is $t = \frac{\hat{\beta}_1 - 10}{SE(\hat{\beta}_1)} \sim t(933)$ under H_0 .

The observed value of the t statistic is $t_{\text{calc}} = \frac{8.3031 - 10}{0.8364} \approx -2.0288$.

The p-value of this test is $\Pr(t(933) > -2.0288) = 0.9786$. Thus, don't reject H_0 both at the 5% and 1% significance levels.

(c) The Confidence Interval at the 99% level is given by $[\hat{\beta}_1 \pm t_{\frac{0.01}{2}}^{(933)} SE(\hat{\beta}_1)]$.

The critical value is $t_{\frac{0.01}{2}}^{(933)} = 2.5811$, thus $CI = [8.3031 \pm 2.1588] = [6.1443, 10.4619]$.

$$(d) \text{wage}_{no} = \hat{\beta}_0 + \hat{\beta}_1(120) = 1113.3636.$$

(e) All calculations here assume A1-A5 from Lecture 3 hold, thus making Thm 3.3 valid.

$$E(\hat{\beta}_0 + 120\hat{\beta}_1 | Q's) = E(\hat{\beta}_0 | Q's) + 120E(\hat{\beta}_1 | Q's) = \beta_0 + 120\beta_1$$

$$\begin{aligned} \text{Var}(\hat{\beta}_0 + 120\hat{\beta}_1 | Q's) &= \text{Var}(\hat{\beta}_0 | Q's) + 120^2 \text{Var}(\hat{\beta}_1 | Q's) + 2(120) \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 | Q's) \\ &= \text{Var}(\hat{\beta}_0 | Q's) + 14,400 \text{Var}(\hat{\beta}_1 | Q's) + 240 \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 | Q's) \end{aligned}$$

Note: since I'm only asking for a formula in the first part, you can leave it like above.

without simplifying, or you can keep going like this: since $s_{xx} = \sum Q_i^2 - (935) \bar{Q}^2 = 211,618.5459$

$$= \frac{\sigma^2}{s_{xx}} \left(\frac{1}{95} \sum Q_i^2 + 14,400 - 240 \bar{Q} \right) = \sigma^2 \left(\frac{576.6765}{211,618.5459} \right) \approx \sigma^2 \times 2.725 \times 10^{-3}$$

→ Note that this is as far as we can go with the formula since we don't know σ^2

An estimate of $E[\hat{\beta}_0 + 120\hat{\beta}_1 | Q's] = \beta_0 + 120\beta_1$ is therefore:

$$\text{wage}_{no} = \hat{\beta}_0 + \hat{\beta}_1(120), \text{ which we calculated as: } 1113.3636 \text{ in (d)}$$

To estimate the variance above, we can use the estimates from $\widehat{\text{Var}}(\hat{\beta} | X's)$, that is:

$$\begin{aligned} \widehat{\text{Var}}(\hat{\beta}_0 + 120\hat{\beta}_1 | X's) &= \widehat{\text{Var}}(\hat{\beta}_0 | X's) + 14,400 \widehat{\text{Var}}(\hat{\beta}_1 | X's) + 240 \widehat{\text{Cov}}(\hat{\beta}_0, \hat{\beta}_1 | X's) \\ &= 7334.47155 + 14,400(0.6996568) + 240(-70.8527594) \\ &= 403.4272 \end{aligned}$$

Note, alternatively if you continued with the formula above, you could also have used the given $\hat{\sigma}^2$ to compute $\hat{\sigma}^2$ which estimates σ^2

(f) We have that $E[\text{wage} | IQ = 120] = \beta_0 + 120\beta_1$. Thus $\widehat{\text{wage}}_{120}$ is an unbiased estimator of the desired predicted wage.

We have also calculated in (d) $\text{Var}(\widehat{\text{wage}}_{120} | IQ's)$, and we also calculated the estimate for this variance $\widehat{\text{Var}}(\widehat{\text{wage}}_{120} | IQ's)$ in the second part.

$$\text{Thus } SE(\widehat{\text{wage}}_{120} | IQ's) = \sqrt{\widehat{\text{Var}}(\widehat{\text{wage}}_{120} | IQ's)} = \sqrt{403.4272} \approx 20.0855$$

Since $(\hat{\beta}_0, \hat{\beta}_1) | X's$ are bivariate normal, and the marginals are also normal by A5, then $\widehat{\text{wage}}_{120}$ is also normally distributed.

With all of the above we can construct the confidence interval as:

$$[\widehat{\text{wage}}_{120} \pm t_{\frac{0.05}{2}}^{(933)} SE(\widehat{\text{wage}}_{120} | IQ's)]$$

Calculating this interval yields: $[1113.3636 \pm 1.96251(20.0855)] = [1073.946, 1152.782]$

$\therefore$ The predicted wage for an individual with $IQ = 120$ has a 95% probability of being in the confidence interval above.